

CSE 150A-250A AI: Probabilistic Methods

Lecture 7

Fall 2025

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Review

Approximate inference in loopy BNs

Rejection sampling

Likelihood weighting

Review

- **Problem**

Given a set E of evidence nodes, and a set Q of query nodes, how to compute the posterior distribution $P(Q|E)$?

- **More precisely**

How to express $P(Q|E)$ in terms of the CPTs $P(X_i|pa(X_i))$ of the BN, which are assumed to be given?

- **Tools at our disposal**

Bayes rule

marginalization

product rule

marginal independence

conditional independence

Exact Inference: Variable Elimination

- **Idea:** Eliminate **redundant** calculations by storing intermediate results in “factors”.
- A **factor** is a function that takes in values of random variables, and produces a number.
- VE works by eliminating all variables in turn until there is a factor with only the query variable.
- To eliminate a variable:
 - *join* all factors containing that variable.
 - *sum* out the influence of the variable on the new factor.

VE Example

$$\begin{aligned}P(J) &= \sum_{M,A,B,E} P(J, M, A, B, E) \\&= \sum_{M,A,B,E} P(J|A)P(M|A)P(B)P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B P(B) \sum_E P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B (P(B)f1(A, B)) \\&= \sum_A P(J|A) \sum_M P(M|A)f2(A) \\&= \sum_A P(J|A)f3(A) \\&= f4(J)\end{aligned}$$

VE Example

$$\begin{aligned}P(J) &= \sum_{M,A,B,E} P(J, M, A, B, E) \\&= \sum_{M,A,B,E} P(J|A)P(M|A)P(B)P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B P(B) \sum_E P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B (P(B)f1(A, B)) \\&= \sum_A P(J|A) \sum_M P(M|A)f2(A) \\&= \sum_A P(J|A)f3(A) \\&= f4(J)\end{aligned}$$

Q. What is the elimination order?

- A. M,A,B,E
- B. E,B,A,M
- C. A,M,B,E

VE Example

$$\begin{aligned}P(J) &= \sum_{M,A,B,E} P(J, M, A, B, E) \\&= \sum_{M,A,B,E} P(J|A)P(M|A)P(B)P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B P(B) \sum_E P(A|B, E)P(E) \\&= \sum_A P(J|A) \sum_M P(M|A) \sum_B (P(B)f1(A, B)) \\&= \sum_A P(J|A) \sum_M P(M|A) f2(A) \\&= \sum_A P(J|A) f3(A) \\&= f4(J)\end{aligned}$$

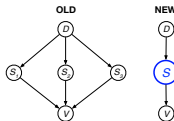
Q. What if we first eliminate M ?

Exact Inference: Variable Elimination

- Time and space of VE is dominated by the **largest** factor created.
 - Can be exponential in the size of BN in the worst case.
- **Heuristic:** Eliminate the variable that will lead to the smallest next factor being created
 - In a **polytree** this leads to **linear** time inference (in size of largest CPT).

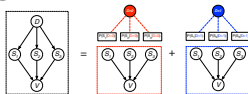
Exact Inference in Loopy BNs

- Transform a loopy BN into a polytree.
- Run the exact inference algorithm.
 - Node Clustering



CPTs grow **exponentially** when nodes are clustered.

- Cutset Conditioning



Number of runs grows **exponentially** with the size of the cutset.

- No efficient algorithm to get clustering or the minimal cutset leads to maximally efficient inference.

Approximate inference in loopy BNs

Motivation

- Formal results

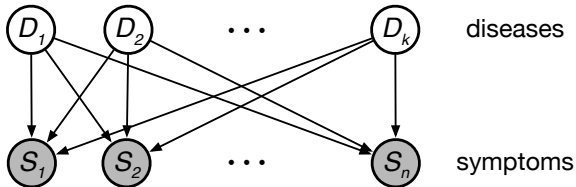
Exact inference in belief networks is **NP-hard**.

Actually it is **#P-hard** (even worse).

- Practical tools

But many large loopy BNs remain useful models.

In these BNs, we must resort to **approximate** methods.



There are many strategies for approximate inference.
We will focus on **stochastic simulation methods**.



This lecture:

1. Prior sampling
2. Rejection sampling
3. Likelihood weighting
4. Markov chain Monte Carlo (MCMC) - Gibbs Sampling

*But before we can describe the above methods,
we must review some basic ideas in sampling ...*

Suppose we have a (biased) coin but we don't know the $P(H)$.

- How can we estimate $P(H)$?
- When will we get the correct probability?

Basic Idea:

- Draw N samples from a sampling distribution S .
- Compute an approximate posterior probability.
- Show that this converges to the true probability P .

We can apply similar intuition for inference in BNs -> Generate samples from a BN.

Sampling a discrete random variable $X \sim P(X)$

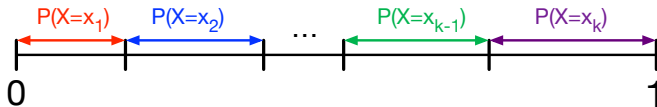
Let X be a discrete random variable with probabilities $P(X=x_i)$. Suppose we can generate random numbers uniformly in $[0, 1)$ (e.g. `random()` in python).

- **Problem**

How to sample values of X so that repeated samples are distributed according to $P(X)$?

- **Solution**

Note that $P(X=x_i)$ defines a partition of unity, which maps a random number $r \in [0, 1)$ into a discrete value of X .



Sampling a discrete random variable $X \sim P(X)$

C	P(C)
orange	0.6
blue	0.1
green	0.3

1. Get sample u from uniform distribution over $[0,1)$.
2. Assign discrete value of C.
 - $0 \leq u < 0.6, C \leftarrow \text{orange}$
 - $0.6 \leq u < 0.7, C \leftarrow \text{blue}$
 - $0.7 \leq u < 1, C \leftarrow \text{green}$

8 samples generated:

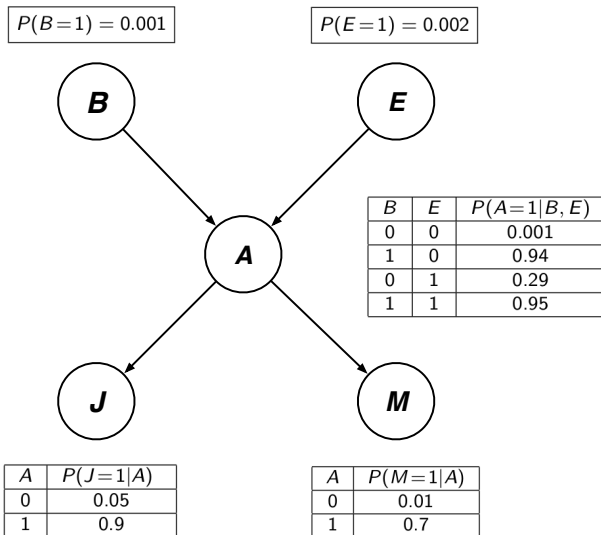
orange, green, orange, blue, orange, orange, green, orange

$$P(C = \text{green})?$$

Belief Network is a **generative** model. We can sample from the distribution represented by the BN.

Idea: Generate one variable at a time in topological order.

Alarm example



Alarm example

- Joint sample

To draw a joint sample $\{b, e, a, j, m\}$ from $P(B, E, A, J, M)$, it is enough to draw the individual samples:

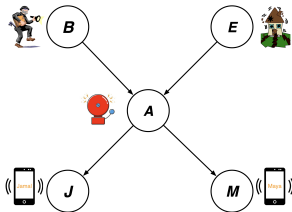
$$b \sim P(B)$$

$$e \sim P(E)$$

$$a \sim P(A|B=b, E=e)$$

$$j \sim P(J|A=a)$$

$$m \sim P(M|A=a)$$



- Convergence in the limit

$$P(b, e, a, j, m) = \lim_{N \rightarrow \infty} \frac{\text{count}(B=b, E=e, A=a, J=j, M=m)}{N}.$$

Prior Sampling from the joint distribution in a discrete BN

Let $\{X_1, X_2, \dots, X_m\}$ be discrete random variables in a BN.

- **Problem**

How to sample values of $\{X_1, X_2, \dots, X_m\}$ so that repeated samples are distributed according to $P(X_1, X_2, \dots, X_m)$?

- **Solution**

The BN defines a **generative model** with joint distribution

$$P(X_1, X_2, \dots, X_m) = \prod_{i=1}^m P(X_i | \text{pa}(X_i)).$$

To draw samples, we can simply take $X_i \sim P(X_i | \text{pa}(X_i))$.

Approximate inference

- Problem

Let Q denote a set of query nodes.

Let E denote a set of evidence nodes.

How can we ~~calculate~~ **estimate** $P(Q|E)$?

- Challenge

While easy to sample from the BN's joint distribution, it may be much harder to sample directly from $P(Q|E)$.

- Solutions

1. rejection sampling (very inefficient)
2. likelihood weighting (more efficient)
3. MCMC (most efficient)

Rejection sampling in BNs

- Problem

Let Q denote a set of query nodes.

Let E denote a set of evidence nodes.

How to estimate $P(Q=q|E=e)$?

- Solution

Generate N samples from the BN's joint distribution.

Discard (or **reject**) the samples where $E \neq e$.

Count the samples $N(q, e)$ where $Q=q$ and $E=e$.

Count the samples $N(e)$ where $E=e$.

Take the ratio of these counts:

$$P(Q=q|E=e) \approx \frac{N(q,e)}{N(e)}$$

where

$$N(q, e) \leq N(e) \leq N$$

Example for rejection sampling

Problem: Estimate $P(a_0|c_1, d_1)$

Samples:

a_1, b_1, c_0, d_0

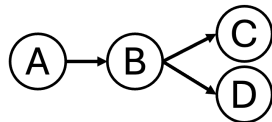
a_0, b_1, c_1, d_0

a_1, b_1, c_0, d_1

a_1, b_1, c_1, d_1

a_1, b_0, c_0, d_0

a_0, b_1, c_1, d_1



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Q. How many samples will be rejected?

A. 6 B. 2 C. 4

D. 3 E. 0

Example for rejection sampling

Problem: Estimate $P(a_0|c_1, d_1)$

Samples:

a_1, b_1, c_0, d_0

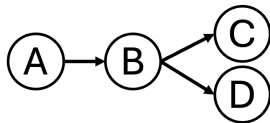
a_0, b_1, c_1, d_0

a_1, b_1, c_0, d_1

a_1, b_1, c_1, d_1

a_1, b_0, c_0, d_0

a_0, b_1, c_1, d_1



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

B	C	P(C B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Q. Estimate of $P(a_0|c_1, d_1)$
using rejection sampling?

A. 1/2 B. 2/3 C. 1/4

D. 1/3 E. 0

Example for rejection sampling

- Sample N times:

$$x_i \sim P(X)$$

$$y_i \sim P(Y|X=x_i)$$

$$e_i \sim P(E|X=x_i)$$

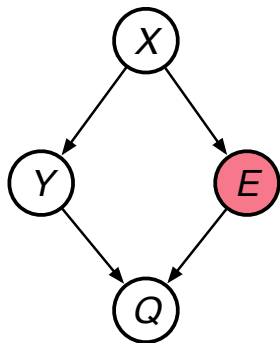
$$q_i \sim P(Q|Y=y_i, E=e_i)$$

- Define the indicator function:

$$I(z, z') = \begin{cases} 1 & \text{if } z=z' \\ 0 & \text{if } z \neq z' \end{cases}$$

- Estimate from ratio:

$$P(Q=q|E=e) \approx \frac{N(q, e)}{N(e)} = \frac{\sum_{i=1}^N I(q, q_i) I(e, e_i)}{\sum_{i=1}^N I(e, e_i)}$$



Properties of rejection sampling

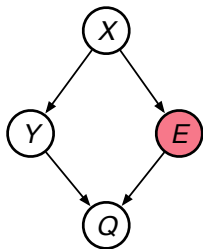
- It converges in the limit:

$$\lim_{N \rightarrow \infty} \frac{N(q, e)}{N(e)} = P(Q=q|E=e)$$

- But it is extremely inefficient:

It discards all samples without $E = e$.

It converges **very slowly** for rare evidence.



How can we do better?

2. Likelihood weighting

- **Key idea**

Instantiate evidence nodes instead of sampling them.

Weight each sample using CPTs at evidence nodes.

- **Intuition**

“Cheat” by fixing the evidence nodes to their desired values.

“Correct” for cheating by penalizing especially unlikely values.

- **Benefits**

No discarding of uninformative samples.

No wasted computation.

Faster convergence.

Example for likelihood weighting

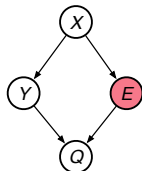
How to estimate $P(Q=q|E=e)$?

- Sample N times:

$$x_i \sim P(X)$$

$$y_i \sim P(Y|X=x_i)$$

$$q_i \sim P(Q|Y=y_i, E=e)$$



Note: E is fixed to e .

- Estimate from ratio:

$$P(Q=q|E=e) \approx \frac{\sum_{i=1}^N l(q, q_i) \overbrace{P(E=e|X=x_i)}^{\text{likelihood weight}}}{\sum_{i=1}^N P(E=e|X=x_i)}$$

- Compare to rejection sampling:

$$P(Q=q|E=e) \approx \frac{\sum_{i=1}^N l(q, q_i) l(e, e_i)}{\sum_{i=1}^N l(e, e_i)}$$

Example with multiple evidence nodes

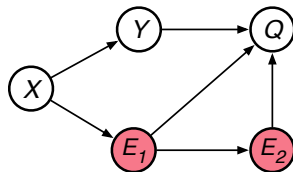
How to estimate $P(Q=q|E_1=e, E_2=e')$?

- Sample N times:

$$x_i \sim P(X)$$

$$y_i \sim P(Y|X=x_i)$$

$$q_i \sim P(Q|Y=y_i, E_1=e, E_2=e')$$



Note: (E_1, E_2) fixed to (e, e')

- Estimate from ratio:

$$P(Q=q|E_1=e, E_2=e') \approx \frac{\sum_{i=1}^N l(q, q_i) \overbrace{P(E_1=e|X=x_i) P(E_2=e'|E_1=e)}^{\text{product of likelihood weights}}}{\sum_{i=1}^N P(E_1=e|X=x_i) P(E_2=e'|E_1=e)}$$

- Compare to rejection sampling:

$$P(Q=q|E=e) \approx \frac{\sum_{i=1}^N l(q, q_i) l(e, e_{1i}) l(e', e_{2i})}{\sum_{i=1}^N l(e, e_{1i}) l(e', e_{2i})}$$

Example for likelihood weighting sampling

Problem: Estimate $P(a_0|c_1, d_1)$

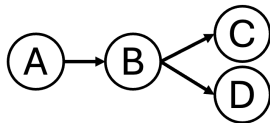
Samples:

a_0, b_1, c_1, d_1

a_1, b_0, c_1, d_1

a_0, b_1, c_1, d_1

Q. Estimate of $P(a_0|c_1, d_1)$
using likelihood weighting?



A	P(A)
a_0	1/5
a_1	4/5

A	B	P(B A)
a_0	b_0	1/4
a_0	b_1	3/4
a_1	b_0	1/3
a_1	b_1	2/3

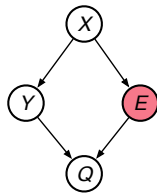
B	C	P(C B)
b_0	c_0	1/5
b_0	c_1	4/5
b_1	c_0	3/5
b_1	c_1	2/5

B	D	P(D B)
b_0	d_0	3/4
b_0	d_1	1/4
b_1	d_0	1/3
b_1	d_1	2/3

Properties of likelihood weighting

- It converges in the limit:

$$\lim_{N \rightarrow \infty} \frac{\sum_{i=1}^N l(q, q_i) P(E=e|X=x_i)}{\sum_{i=1}^N P(E=e|X=x_i)} = P(Q=q|E=e)$$

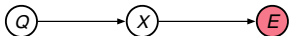


- It's more efficient than rejection sampling:

No samples need to be discarded.

Descendants of evidence nodes are conditioned on evidence.

- But it can still be very slow:



The worst case for likelihood weighting is when rare evidence is descended from query nodes.

That's all folks!